

Common BIOS and POST Error Messages

No	BIOS/POST Error Message	Usually Indicates
1	CMOS Checksum Error	CMOS battery problem or incorrect/corrupted BIOS settings.
2	Keyboard Error	keyboard may not be connected properly or may have a hardware problem.
3	No Boot Device Found	The storage drive may not be detected or the boot device/boot order may be incorrect.
4	Memory Error	RAM may be loose, faulty, or incompatible.
5	CPU Fan Error	The CPU cooling fan may not be connected correctly or may not be working properly.

conclusion:

BIOS and POST errors help identify hardware or configuration problems during the computer startup

process. Checking the displayed message or beep pattern can help locate the possible problem.

Troubleshooting Three POST Beeps

When the PC starts, it produces 3 beeps repeatedly and there is no display on the screen.

Meaning:

For an **AMI BIOS**, 3 beeps typically indicate a **memory/RAM-related problem**. The exact beep-code meaning can vary by BIOS version and motherboard, so it should be confirmed with the motherboard documentation.

Troubleshooting Step 1:

I would switch off the PC, disconnect the power, and check whether the RAM modules are properly seated in their slots.

I would remove and reinstall them carefully.

Troubleshooting Step 2:

I would test the RAM modules one at a time in the recommended slot to identify whether one module or slot may be causing the problem.

Conclusion:

Since there is no display and the system gives 3 repeating

beeps, I would first focus on checking the RAM and its connection.

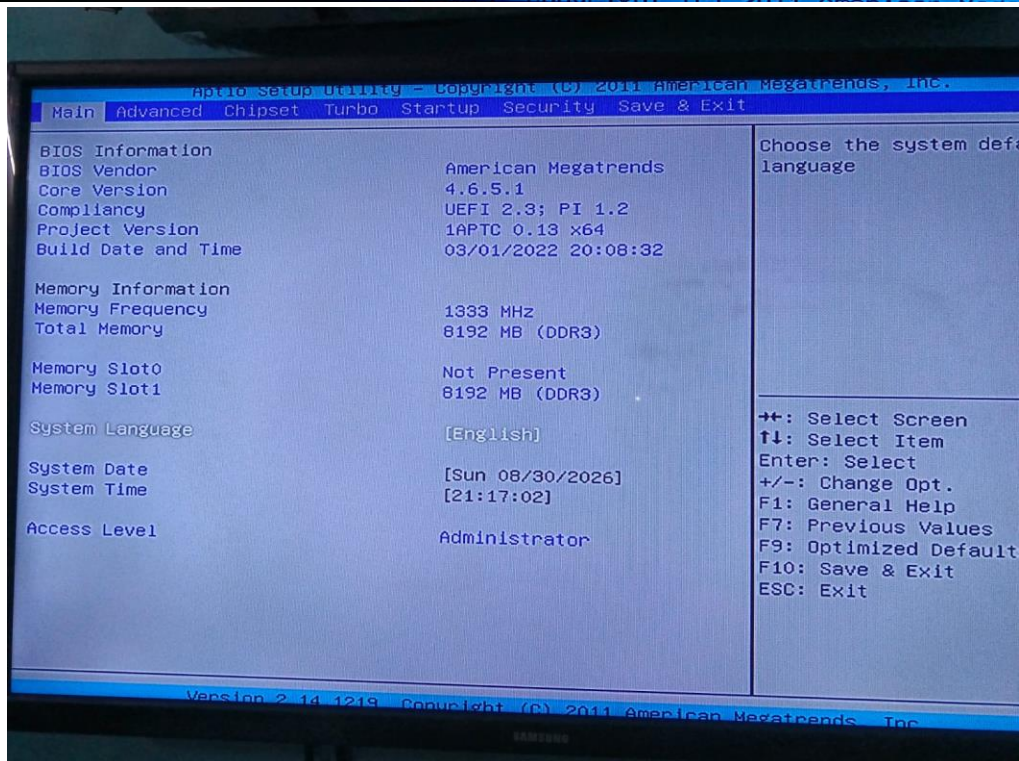
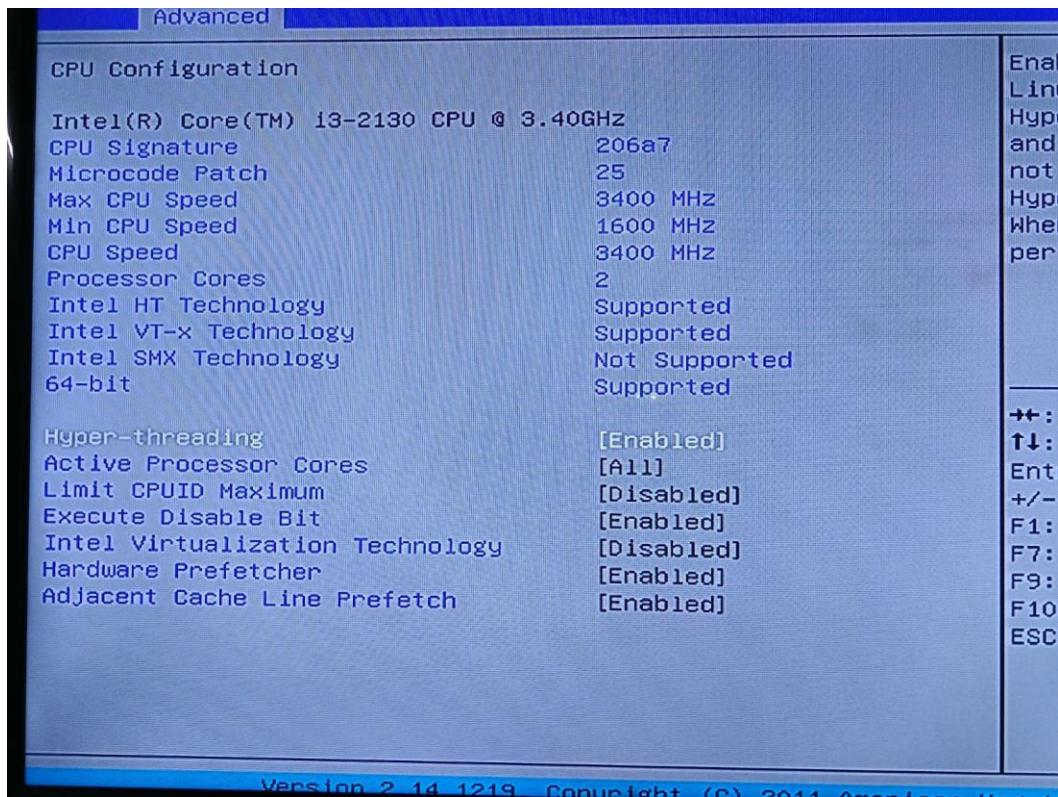
Accessing BIOS Setup Utility

I accessed the BIOS setup utility and checked the main BIOS screen. I identified the motherboard manufacturer and the BIOS version.

BIOS Information:

- BIOS Vendor: American Megatrends
- Core Version: 4.6.5.1
- BIOS/UEFI: UEFI 2.3; PI 1.2
- Project Version: 1APTC 0.13 x64

- Build Date: 03/01/2022
- Total Memory: 8192 MB (DDR3)
- MB Manufacturer: American Megatrends
- BIOS Version: Core Version 4.6.5.1



CMOS Checksum Error

The message “**CMOS checksum error – Defaults loaded**” means that the BIOS could not verify the stored CMOS settings correctly. As a result, the system has loaded the default BIOS settings.

Possible Cause:

A weak or dead **CMOS battery** is one common cause. Incorrect or corrupted BIOS settings can also cause this message.

How I Would Resolve It:

1. I would restart the PC and enter the BIOS setup.
2. I would check whether the date, time, and other BIOS settings are correct.
3. I would load the recommended/default BIOS settings if required.
4. If the error continues, I would replace the CMOS battery with the correct type.
5. After replacing the battery, I would enter BIOS again and set the date, time and required settings.
6. Finally, I would restart the PC and check whether the error has been resolved.

Conclusion:

The CMOS checksum error usually indicates that the stored BIOS settings are not being retained correctly.

Checking the BIOS settings and replacing a weak CMOS battery can help resolve the problem.